

MECANISMO-PISTON-BIELA-MANIVELA-...



laktecuento



Cinemática y Dinámica de Máquinas



3º Grado en Ingeniería Mecánica



Escuela de Ingeniería de Barcelona este (Eebe)
Universidad Politécnica de Catalunya



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(Estudia y no nos des ideas)

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1/6

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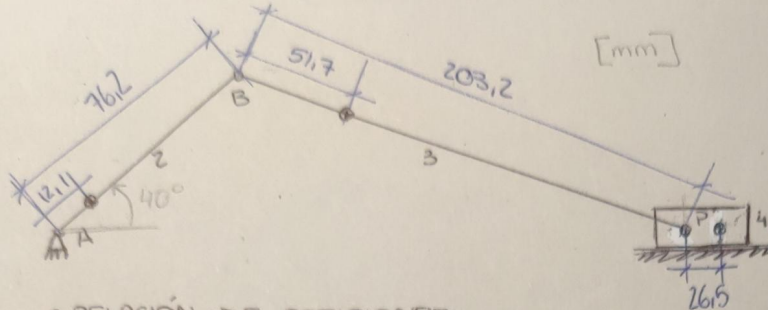


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MECANISMO DE PISTÓN-BIELA-MANIVELA DE UN MCIA



$$\varphi_2 = 40^\circ$$

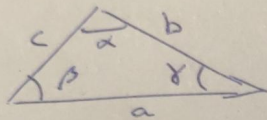
$$\omega_2 = 2000 \text{ rpm} \cdot \frac{2\pi \text{ rad}}{60 \text{ s}} = \frac{200}{3} \pi \frac{\text{rad}}{\text{s}}$$

$$\alpha_2 = 20.9 \text{ rad/s}^2$$

• RELACIÓN DE POSICIONES:

TEOREMA DEL SENOS:

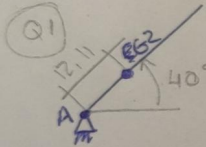
$$\frac{203.2}{\sin(40^\circ)} = \frac{76.2}{\sin(\gamma)} \rightarrow \gamma = \varphi_3 = 13.9483^\circ$$



$$\vec{q} = \{40, 13.9483\}^\circ \text{ deg}$$

• VELOCIDADES Y ACELERACIONES EN LOS CG'S:

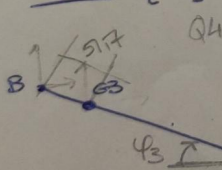
DSL 2 [RP] $\vec{V}_{G2} = \vec{\omega}_2 \times \vec{r}_{G2A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \frac{200\pi}{3} \\ 0.1211\text{m} & 0.1215\text{m} & 0 \end{vmatrix} = -16.303 \hat{i} + 19.429 \hat{j} \frac{\text{m}}{\text{s}} = \vec{V}_{G2}$



$$\vec{a}_{G2} = \vec{\alpha}_2 \times \vec{r}_{G2A} + \vec{\omega}_2 \times (\vec{\omega}_2 \times \vec{r}_{G2A})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 20.9 \\ 0.1211\text{m} & 0.1215\text{m} & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \frac{200\pi}{3} \\ -16.303 & 19.429 & 0 \end{vmatrix} = -407.0927 \hat{i} - 341.2553 \hat{j} \frac{\text{m}}{\text{s}^2} = \vec{a}_{G2}$$

DSL 3 [MG]



$$\vec{V}_{G3} = \vec{V}_B + \vec{\omega}_3 \times \vec{r}_{G3B}$$

$$\vec{V}_B = \vec{V}_{G2} \cdot \frac{0.0762}{0.01211} \rightarrow -10.2584 \hat{i} + 12.2254 \hat{j} \frac{\text{m}}{\text{s}} = \vec{V}_B$$

IR A ELEMENTO 4

$$\vec{a}_{G3} = \vec{a}_B + \alpha_3 \times \vec{r}_{G3B} + \omega_3 \times (\omega_3 \times \vec{r}_{G3B})$$

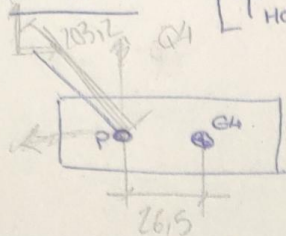
$$\vec{a}_B = \vec{a}_{G2} \cdot \frac{0.0762}{0.01211} \rightarrow -2561.4948 \hat{i} - 2147.2877 \hat{j} \frac{\text{m}}{\text{s}^2} = \vec{a}_B$$

WUOLAH

DSL 4

 $[T_{HORIZ}]$

$$V_P = V_{G4}$$



$$\varphi_3 = 13.9483^\circ$$

$$V_P = V_B + \vec{\omega}_3 \times \vec{r}_{PB}$$

$$= (-10.2584 \hat{i} + 12.2254 \hat{j}) + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega_3 \\ +0.2032 C_{\varphi_3} & -0.2032 S_{\varphi_3} & 0 \end{vmatrix} \rightarrow$$

$$\rightarrow -10.2584 \hat{i} + 12.2254 \hat{j} + (0.04898 \omega_3 \hat{i} + 0.1972 \omega_3 \hat{j}) = V_P \hat{i} \rightarrow$$

$$\hat{i} : V_P = -10.2584 + 0.04898 \omega_3 \rightarrow V_P = -13.2949 \text{ m/s } \hat{i}$$

$$\hat{j} : 0 = 12.2254 + 0.1972 \omega_3 \rightarrow \omega_3 = -61.9949 \text{ rad/s}$$

$$V_{G4} = -13.2949 \hat{i} + 0 \hat{j} \text{ m/s}$$

$$\vec{V}_{G3} = \vec{V}_B + \vec{\omega}_3 \times \vec{r}_{G3B} = (-10.2584 \hat{i} + 12.2254 \hat{j}) + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -61.9949 \\ 0.0517 C_{\varphi_3} & -0.0517 S_{\varphi_3} & 0 \end{vmatrix} =$$

$$V_{G3} = -11.031 \hat{i} + 9.1148 \hat{j} \text{ m/s}$$

$$A_{G4} = A_P$$

$$\vec{A}_P = \vec{A}_B + \vec{\alpha}_3 \times \vec{r}_{PB} + \vec{\omega}_3 \times (\vec{\omega}_3 \times \vec{r}_{PB})$$

$$= \underbrace{-2561.4948 \hat{i} - 2147.2877 \hat{j}}_{(1)} + \underbrace{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \alpha_3 \\ 0.2032 C_{\varphi_3} & -0.2032 S_{\varphi_3} & 0 \end{vmatrix}}_{(2)} + \underbrace{(-61.9949 \hat{k}) \times \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -61.9949 \\ 0.2032 C_{\varphi_3} & -0.2032 S_{\varphi_3} & 0 \end{vmatrix}}_{(3)} =$$

$$\vec{A}_P \hat{i} = (-2561.4948 \hat{i} - 2147.2877 \hat{j}) + (0.04898 \alpha_3 \hat{i} + 0.1972 \alpha_3 \hat{j}) + (-757.9443 \hat{i} + 188.2505 \hat{j}) \rightarrow$$

$$\hat{i} : A_P = -2561.4948 + 0.04898 \alpha_3 - 757.9443 \rightarrow A_P = -2832.8588 \text{ m/s}^2 \hat{i}$$

$$\hat{j} : 0 = -2147.2877 + 0.1972 \alpha_3 + 188.2505 \rightarrow \alpha_3 = 9934.2657 \frac{\text{rad}}{\text{s}^2}$$

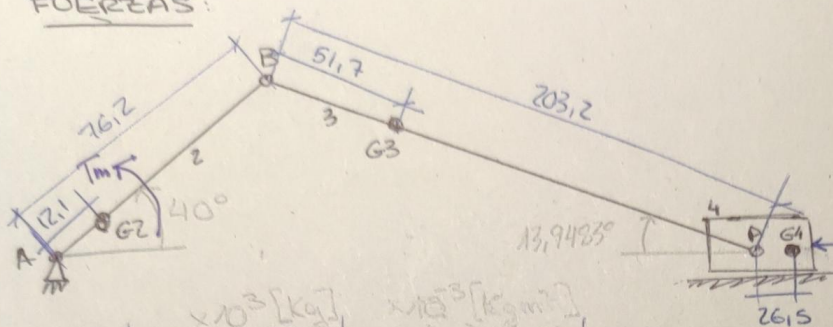
$$A_{G4} = -2832.8588 \hat{i} + 0 \hat{j} \text{ m/s}^2$$

$$A_{G3} = \vec{A}_B + \vec{\alpha}_3 \times \vec{r}_{G3B} + \vec{\omega}_3 \times (\vec{\omega}_3 \times \vec{r}_{G3B})$$

$$= \vec{A}_B + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 9934.2657 \\ 0.0517 C_{\varphi_3} & -0.0517 S_{\varphi_3} & 0 \end{vmatrix} + (-61.9949 \hat{k}) \times \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -61.9949 \\ 0.0517 C_{\varphi_3} & -0.0517 S_{\varphi_3} & 0 \end{vmatrix} =$$

$$A_{G3} = -2630.5362 \hat{i} - 1600.934 \hat{j} \text{ m/s}^2$$

FUERZAS:



$$\mu = 0.015$$

$$F_P = 1500 \text{ N} \leftarrow$$

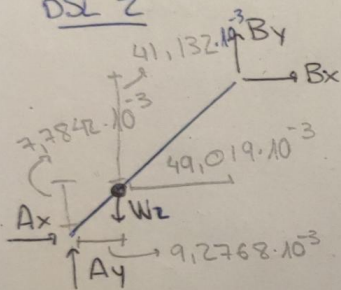
$$\omega_2 = 2000 \text{ rpm} = \frac{2000\pi}{3} \text{ rad/s}$$

$$\alpha_2 = 20.9 \text{ rad/s}^2$$

$$\varphi_2 = 40^\circ$$

#	$m \text{ (TON)} \times 10^3 [\text{kg}]$	$I_G \text{ (TON} \cdot \text{mm}^2) \times 10^5 [\text{kg} \cdot \text{m}^2]$	$V_G \text{ (m/s)}$	$A_G \text{ (m/s}^2)$	$\omega \text{ (rad/s)}$	$\alpha \text{ (rad/s}^2)$
2	$6.9838 \cdot 10^{-5}$	$8.3242 \cdot 10^{-2}$	$(-1.6303, 1.9429)$	$(-407.0827, -341.2663)$	$\frac{2000\pi}{3}$	20.9
3	$4.6232 \cdot 10^{-4}$	2.9307	$(-11.031, 9.1148)$	$(2630.5362, -1600.934)$	-61.9949	9934.2657
4	$5.1563 \cdot 10^{-3}$	4.2571	$(-13.2949, 0)$	$(-2832.8588, 0)$	0	0

DSL 2

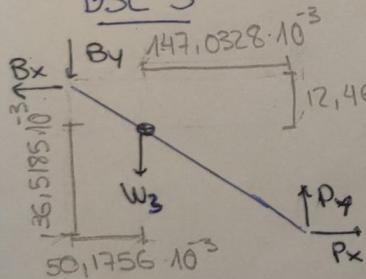


$$\begin{cases} \sum F_x - m_k \cdot a_{Gxk} = 0 \rightarrow A_x + B_x - m_2 \cdot a_{G2x} = 0 & [1] \\ \sum F_y - m_k \cdot a_{Gyk} = 0 \rightarrow A_y + B_y - W_2 - m_2 \cdot a_{G2y} = 0 & [2] \\ \sum M_G - I_{Gk} \alpha_k = 0 \rightarrow T_m + 7.7842 \cdot 10^{-3} A_x - 9.2763 \cdot 10^{-3} A_y + 49.019 \cdot 10^{-3} B_y - 41.132 \cdot 10^{-3} B_x - 8.3242 \cdot 10^{-5} \cdot 20.9 = 0 & [3] \end{cases}$$

A, B, Tm
x2 x2

7 INC / 6 ECS

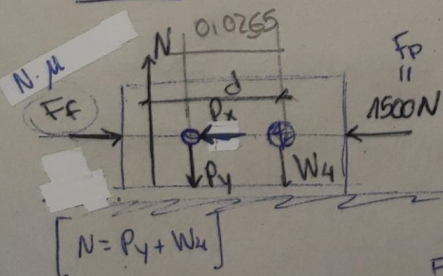
DSL 3



$$\begin{cases} -B_x + P_x - m_3 \cdot a_{G3x} = 0 & [4] \\ -B_y + P_y - W_3 - m_3 \cdot a_{G3y} = 0 & [5] \\ 12.4621 \cdot 10^{-3} B_x + 50.1756 \cdot 10^{-3} B_y + 147.0328 \cdot 10^{-3} P_y + 36.5185 \cdot 10^{-3} P_x - 2.9307 \cdot 10^{-3} \cdot 9934.2657 = 0 & [6] \end{cases}$$

A, B, P, Tm
x2 x2 x2

DSL 4



$$(P_y + W_4) \cdot \mu - P_x - F_P - m_4 \cdot a_{G4x} = 0 \quad [7]$$

$$-P_y - W_4 + N - m_4 \cdot a_{G4y} = 0$$

$$0.0265 \cdot P_y - 4.2571 \cdot 0 = 0$$

7 INC / 7 ECS

$$F_f = N \cdot \mu = (P_y + W_4) \cdot \mu$$



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$$\begin{cases} A_x + B_x = -0.1122 & [1] \\ A_y + B_y = 0.809 & [2] \\ 7.7842 \cdot 10^{-3} A_x - 9.2763 \cdot 10^{-3} A_y - 41.132 \cdot 10^{-3} B_x + 49.019 \cdot 10^{-3} B_y + T_m = 1.7398 \cdot 10^{-3} & [3] \\ -B_x + P_x = -5.0999 & [4] \\ -B_y + P_y = 8.7493 & [5] \\ 12.4621 \cdot 10^{-3} B_x + 50.1756 \cdot 10^{-3} B_y + 36.5185 \cdot 10^{-3} P_x + 147.0328 \cdot 10^{-3} P_y = 29.1144 & [6] \\ -P_x + 0.015 P_y = -13106.3111 & [7] \end{cases}$$

A_x	A_y	B_x	B_y	P_x	P_y	T_m		
1	0	1	0	0	0	0	A_x	-0.1122
0	1	0	1	0	0	0	A_y	0.809
$7.7842 \cdot 10^{-3}$	$-9.2763 \cdot 10^{-3}$	$-41.132 \cdot 10^{-3}$	$49.019 \cdot 10^{-3}$	0	0	1	B_x	$1.7398 \cdot 10^{-3}$
0	0	-1	0	1	0	0	B_y	-5.0999
0	0	0	-1	0	1	0	P_x	8.7493
0	0	$12.4621 \cdot 10^{-3}$	$50.1756 \cdot 10^{-3}$	$36.5185 \cdot 10^{-3}$	$147.0328 \cdot 10^{-3}$	0	P_y	29.1144
0	0	0	0	-1	0.015	0	T_m	-13106.3111

$$A_x = -13065.111 \text{ N}$$

$$A_y = 3103.7056 \text{ N}$$

$$B_x = 13064.9988 \text{ N}$$

$$B_y = -3102.8966 \text{ N}$$

$$P_x = 13059.8989 \text{ N}$$

$$P_y = -3094.1473 \text{ N}$$

$$T_m = 819.9845 \text{ Nm}$$

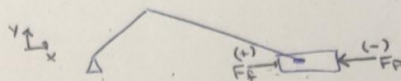
MATLAB

EL CÁLCULO COMO TAL ESTÁ MAL YA QUE HAY 2 ELEMENTOS EN EL SÓLIDO 2. EL PROFESOR Dijo QUE NO LOS CONSIDERARAMOS. POR LA RAZÓN QUE ACABO DE DECIR.

EL PESO DE [EL PESO 2]
EL RESULTADO SOLO CONSIDERA UNA DE LAS BARRAS, POR ESO ESTÁ MAL EL RESULTADO FINAL, PERO, EL PROCEDIMIENTO "ES CORRECTO".

WUOLAH

POTENCIAS VIRTUALES



FORMULA:
$$\underbrace{\sum_{i=1} \vec{F}_i \cdot \vec{V}_i}_{W_2, W_3, W_4, F_p, F_f} + \underbrace{\sum_{j=1} \vec{M}_j \cdot \vec{\omega}_j}_{T_m} - \underbrace{\sum_{k=1} m_k \cdot \vec{a}_{Gk} \cdot \vec{V}_{Gk}}_{m_2, m_3, m_4} - \underbrace{\sum_{k=1} I_{Gk} \cdot \vec{\alpha}_k \cdot \vec{\omega}_k}_{I_{G2}, I_{G3}, I_{G4}} = 0$$

(I) (II) (III) (IV)

RESOLUCIÓN
MEDIANTE
EXCEL

(I) $\sum \vec{F}_i \cdot \vec{V}_i =$

$= W_2 \cdot V_{G2} + W_3 \cdot V_{G3} + W_4 \cdot V_{G4} + F_p \cdot V_{G4} + F_f \cdot V_{G4} =$

$= m_2 \cdot g \cdot V_{G2y} + m_3 \cdot g \cdot V_{G3y} + m_4 \cdot g \cdot V_{G4y} - F_p \cdot V_{G4x} + F_f \cdot V_{G4x} =$

$= 1,312 + 41,3389 + 0 + 19942,35 - 627,1332 \rightarrow \boxed{\sum \vec{F}_i \cdot \vec{V}_i = 19357,8678}$

(II) $\sum \vec{M}_j \cdot \vec{\omega}_j =$

$= \vec{T}_m \cdot \vec{\omega}_2 = T_m \cdot \omega_2 = \boxed{209,4395 \cdot T_m = \sum \vec{M}_j \cdot \vec{\omega}_j}$

(III) $\sum m_k \cdot \vec{a}_{Gk} \cdot \vec{V}_{Gk} =$

$= m_k \cdot [(a_{Gkx} \hat{i} + a_{Gky} \hat{j}) (V_{Gkx} \hat{i} + V_{Gky} \hat{j})] = m_k \cdot (a_{Gkx} \cdot V_{Gkx} + a_{Gky} \cdot V_{Gky}) \Rightarrow$

$\Rightarrow 0,10442 + 6669,0823 + 194199,5327 \Rightarrow \boxed{\sum m_k \cdot \vec{a}_{Gk} \cdot \vec{V}_{Gk} = 200868,6592}$

(IV) $\sum I_{Gk} \cdot \vec{\alpha}_k \cdot \vec{\omega}_k =$

$= I_{G2} \cdot \alpha_2 \cdot \omega_2 + I_{G3} \cdot \alpha_3 \cdot \omega_3 + I_{G4} \cdot \alpha_4 \cdot \omega_4$

$= 0,3644 - 1804,9414 + 0 \Rightarrow \boxed{\sum I_{Gk} \cdot \vec{\alpha}_k \cdot \vec{\omega}_k = -1804,577}$

Tq.

$19357,8678 + 209,4395 T_m - (200868,6592) - (-18049413,3456) = 0$

$T_m = \frac{-19357,8678 + 200868,6592 - 1804,577}{209,4395}$

$\boxed{T_m = 858,03 \text{ Nm}}$ $\rightarrow$ LO MISMO PARA
AQUÍ.